

Planning and measurement

Slide 1 · 0–2 min

Planning and measurement

Every technique must discharge a named verification claim.

Plan the claim → Model what is sampled → Close holes honestly

Teaching script

Open with a decision the audience must make in this area. Ask which result they currently treat as sufficient evidence and keep that answer for the closing discussion.

The lesson selects a practical argument from the book. The study handout preserves the detailed examples, citations and worked artifacts.

Source: ch05:13–65, ch06:10–36, ch06:389–449, ch07:9–20

Learning outcomes

Slide 2 · 2–5 min

Learning outcomes

Demonstrate these decisions on a real plan row.

- Write a plan row with one closure criterion.
- Design sampling around the event of interest.
- Classify a hole with a reviewable reason.
- Explain a readiness decision using correlated evidence.

Teaching script

Read the outcomes as observable tasks. Ask learners to identify which one they can already demonstrate and which one currently depends on undocumented judgment.

At closure, require a concrete artifact or explanation for an outcome rather than a general expression of confidence.

Source: ch05:13–65, ch06:10–36, ch06:389–449, ch07:9–20

Route through the lesson

Slide 3 · 5–7 min

Route through the lesson

Four blocks, each ending with a decision.

- Plan the claim
- Model what is sampled
- Close holes honestly
- Read the evidence together

Teaching script

Allow eighteen minutes for each block: a concept, its engineering consequence, a worked interpretation and a short discussion. Keep the discussion tied to the evidence available in the example.

Reserve the final eleven minutes for the assigned exercise and application to the learners' work. The timing is a course design allocation, not an engineering estimate.

Source: ch05:13–65, ch06:10–36, ch06:389–449, ch07:9–20

Extract behavior, not headings

Slide 4 · 7–11 min

Extract behavior, not headings

Omissions in the plan become omissions in verification.

- Read interfaces, functions and architecture.
- Record forbidden and out-of-scope behavior.

Teaching script

Walk a DMA description from register interface to transfer behavior and then to architectural interactions. A section heading is not yet an independently checkable feature.

Ask what the designer knows about corner cases that the prose does not say. Keep the must-not-happen list explicit; a happy-path feature list can be complete in appearance and still omit the escape.

Source: ch05:13–65

A row needs one closure argument

Slide 5 · 11–16 min

A row needs one closure argument

Separate claims that require different evidence.

Feature and attributes → Pass/fail criterion → Method and closure

Teaching script

Use a protocol property and a functional transfer result to illustrate why one feature description can need separate rows. The checker and closure metric must match the kind of wrong behavior.

Require the criterion before choosing the method. A test name or the phrase fully verified does not specify what will be observed or what discharges the row.

Source: ch05:66–152

Keep traceability executable

Slide 6 · 16–21 min

Keep traceability executable

Stable identifiers connect intent to results.

- Link specification, feature, tests and coverage.
- Make missing links visible in review.

Teaching script

Trace one permanent feature ID into the regression selection and back from its result. The direction matters: the report must identify which claim was checked, not merely which executable ran.

Invite design, systems, test and outside peers to challenge correctness, precision and completeness. Explain that a tool can preserve links but cannot detect an important feature nobody entered.

Source: ch05:153–162, ch05:163–180

Discussion: one row, two verdicts

Slide 7 · 21–25 min

Discussion: one row, two verdicts

Name the evidence that would settle the decision.

- A row claims legal bursts and correct copied data.
- Its only criterion is an assertion result.
- How should the plan change?

Teaching script

Expected answer: separate the protocol-legality claim from the data-correctness claim, assigning the appropriate property and functional comparison.

The distinction is evidence, not bureaucracy. A single passing protocol result cannot discharge a different data claim merely because both concern the DMA.

Source: ch05:66–152

Coverage families have different subjects

Slide 8 · 25-29 min

Coverage families have different subjects

A full structural report can miss absent functionality.

Implementation → Specification → Checker activity

Teaching script

Ask where each model comes from. Code coverage is derived from implementation structure; functional coverage is authored from intended behavior; assertion activity describes whether checkers were exercised.

An omitted implementation cannot contribute code to cover. Use that observation to explain why structural completeness and specification completeness are different review targets.

Source: ch06:10-36, ch06:37-72

Bins belong at behavior boundaries

Slide 9 · 29–34 min

Bins belong at behavior boundaries

Convenient partitions can merge cases that behave differently.

- Choose attributes from the specification.
- Shape crosses to justified combinations.
- Sample attributes when they describe the same event.

Teaching script

Use the accelerator example: width, precision and contention matter because they select different behavior. Equal-sized automatic bins need not respect those boundaries.

Ask when the attributes become jointly valid. A correct list of attributes sampled at unrelated times produces a misleading model even when its report looks complete.

Source: ch06:73–91, ch06:109–223

Sampling is not checking

Slide 10 · 34–39 min

Sampling is not checking

A filled bin needs an active correctness oracle.

- Coverage records that a modeled situation occurred.
- A checker determines whether its behavior was acceptable.

Teaching script

Present a campaign optimized for bin hits. Ask what connects each hit to a mechanism that would reject a wrong result. The optimization target can be satisfied while that mechanism is disconnected.

Require coverage-model review to include the observing checkers. Do not replace this review with a comparison count: activity is necessary evidence, not a proof of checker correctness.

Source: ch06:92–108

Discussion: contention at the wrong instant

Slide 11 · 39–43 min

Discussion: contention at the wrong instant

Name the evidence that would settle the decision.

- Contention is sampled only when a job retires.
- The defect requires contention during streaming.
- What must the model remember?

Teaching script

Expected answer: latch or otherwise preserve whether contention occurred during the relevant streaming window, then sample attributes together at a defined completion event.

Explain why the same report can be produced by a good and bad sampling model. The semantic error lives in the relation between the recorded attributes and the event of interest.

Source: ch06:109–223, ch06:224–330

A coverage hole is a diagnosis

Slide 12 · 43–47 min

A coverage hole is a diagnosis

Different causes require different actions.

- Check stimulus reachability and model intent.
- Distinguish unexercised behavior from impossible behavior.

Teaching script

Follow a hole through classification before proposing more seeds. A model mistake, a generator restriction and a rare reachable case require different fixes.

Keep formal unreachability claims tied to their assumptions. A constraint in the stimulus library is not automatically a restriction of the design's legal behavior.

Source: ch06:389–449

An exclusion is a reviewable artifact

Slide 13 · 47–52 min

An exclusion is a reviewable artifact

The report must retain the reason a hole was removed.

- Classify the exclusion.
- Record reason, owner and relevant version.
- Review it when the design or plan changes.

Teaching script

Distinguish structural restrictions, specification restrictions and accepted risk. Each carries a different justification and authority.

Ask which change would invalidate the reason. A waiver copied from a previous configuration is an assumption that needs re-examination, not permanent evidence.

Source: ch06:450–541

Risk determines depth

Slide 14 · 52–57 min

Risk determines depth

Good observability makes checking affordable; it does not remove risk.

- Consider consequence, novelty and specification quality.
- Agree priority and cut lines before the schedule crisis.

Teaching script

Contrast the new DMA stride path with generated, well-specified register behavior. The chapter spends verification effort according to escape consequence and likelihood.

When a specification is incomplete, record zones and explicit defects. A local interpretation embedded in code must not become an unreviewed requirement merely because implementation has begun.

Source: ch05:181–206, ch05:207–278

Discussion: unreachable or just ungenerated?

Slide 15 · 57–61 min

Discussion: unreachable or just ungenerated?

Name the evidence that would settle the decision.

- A legal bin has never been hit.
- The sequence library does not generate it.
- Is a structural exclusion justified?

Teaching script

Expected answer: no. The observation establishes a stimulus limitation, not structural impossibility. Either expand stimulus or record a separately justified risk decision.

Ask what would justify unreachability: a specification restriction or a proof under reviewed assumptions. Have learners state the difference in the exclusion's reason field.

Source: ch06:389–449, ch06:450–541

A dashboard is a set of instruments

Slide 16 · 61–65 min

A dashboard is a set of instruments

No single curve determines readiness.

- Read per-feature status with coverage trends.
- Correlate bug curves, regression health and debug effort.

Teaching script

Treat the plan as the organizing measure and use the other instruments to challenge its interpretation. A dashboard should trigger a decision, not simply decorate a status report.

Ask whether the same apparent improvement could be caused by reduced stimulus, disabled checking or a changing denominator. State the companion measurement that would discriminate.

Source: ch07:9–20, ch07:21–38

A flat bug curve is ambiguous

Slide 17 · 65–70 min

A flat bug curve is ambiguous

Clean behavior and stimulus starvation can look alike.

- Check freshness and breadth of stimulus.
- Inspect checker integrity and where recent bugs were found.

Teaching script

Offer two explanations for flat discovery and ask the room to distinguish them using available data. The chapter requires fresh-stimulus and coverage information rather than inferring quality from the curve alone.

A late severe bug in an unmodeled region changes the model and the plan. Demonstrate stability again under the expanded instrumentation; the old flat curve did not watch that region.

Source: ch07:39–67

Sign-off is an evidence decision

Slide 18 · 70–75 min

Sign-off is an evidence decision

Conditional release must name the accepted gap.

- Discharge plan rows and review exclusions.
- Record reproducible baselines and bug dispositions.
- A conditional decision needs an owner and expiry.

Teaching script

Use the chapter's three possible outcomes: signed off, conditional or held. Ask learners to distinguish a conditional decision from an informal promise to fix something later.

The waiver names the hole, reason, risk argument, owner and expiry. The record must permit someone outside the meeting to reconstruct what was accepted.

Source: ch07:68–116

Discussion: a late severe bug

Slide 19 · 75–79 min

Discussion: a late severe bug

Name the evidence that would settle the decision.

- Discovery had been flat.
- A new checker finds a severe unmodeled failure.
- Which earlier conclusion must be revisited?

Teaching script

Expected answer: the old stability evidence did not observe the newly discovered region. Update the plan, coverage and checking as necessary, then establish the required evidence under the changed instrumentation.

Do not treat fixing the immediate RTL defect as completing the escape analysis. The process that allowed the missing observation must also produce an upstream change.

Source: ch07:39–67, ch07:145–225

Exercise: coverage without comparisons

Slide 20 · 79–85 min

Exercise: coverage without comparisons

Decide what the report can establish.

- The modeled space is fully hit.
- No output comparison was performed.
- State the missing evidence and a defensible closure decision.

Teaching script

Exercise 6.3 (judgement). A closure tool reads the uncovered bins back from a merged database, converts them mechanically into constraints, and brings the accelerator's model from a plateau to full coverage overnight. Decide what that result establishes about the campaign and what it does not, and name the property of coverage closure §6.4 states that makes the distinction necessary. List the stopping criteria of §6.8 the result does not by itself satisfy, and give the reason §6.8 offers for tightening the checking review when closure is automated.

Use book Exercise 6.3. Give learners time individually, then compare answers in pairs before the debrief.

Expected answer: the report measures sampling across an authored model, not correctness or completeness of that model. Closure needs a reviewed full model, current exclusions and stability under refinement, together with active checking of behavior. Reject an unconditional closure based on bin hits alone.

Source: ch06:92–108

What to change on Monday

Slide 21 · 85–87 min

What to change on Monday

Make one existing verification decision reviewable.

- Review one row with mixed closure criteria.
- Trace a coverage hit to its checker.
- Challenge one exclusion or dashboard inference.

Teaching script

Ask each learner to choose one of these actions in an existing project. Require a small, inspectable deliverable and identify the person who will review it.

Keep the change connected to the lesson's evidence limits. A new dashboard, script or model is useful only if it strengthens the claim the plan row is meant to establish.

Source: ch05:13–65, ch06:10–36, ch06:389–449, ch07:9–20

Scope and next reading

Slide 22 · 87–90 min

Scope and next reading

Use the handout for the detailed artifacts and sources.

- Read the Planning and measurement handout and book Chapters 5–7.
- Tool configuration and full coverage syntax are outside this lesson.

Teaching script

Revisit the outcomes and ask the room to state a limitation of the strongest result discussed today. A complete answer identifies a model boundary, an assumption, a missing oracle or the scope of the underlying evidence.

Assign the detailed reading to the relevant chapters. The omitted topics are explicit scope boundaries, not claims that those topics are unimportant.

Source: ch05:13–65, ch06:10–36, ch06:389–449, ch07:9–20